
Hidden Consensus, Loud Disagreement: When the ML Researcher Is an AI

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Abstract

1 When AI agents, standing in for the researcher, are asked to answer the same
2 research question from the same data, do they agree on what they find? We test
3 this for machine learning specifically, where results are directly checkable by
4 execution. Across 240 independent, memory-isolated agent replicates analyzing
5 the UCI Adult/Census Income dataset under six research questions, we measure
6 *measure-choice diversity*: how many distinct ways replicates choose to measure
7 the same answer. This turns out to depend heavily on the specific question, not
8 simply on whether it is worded abstractly or concretely. One abstract question
9 alone produced 10 distinct model-comparison choices out of 20 replicates; two of
10 the other three abstract questions, by contrast, converge tightly on a single measure
11 once cosmetic wording differences are collapsed. Pooled across questions, abstract
12 framing still roughly doubles diversity over concrete framing (2.6 vs. 1.3 distinct
13 measures per 10 replicates), but the effect is concentrated, not uniform. Despite
14 this surface disagreement, replicates converged on the same qualitative conclusion
15 in 75–100% of cases on two abstract questions, even while disagreeing widely on
16 which number to report. Loud disagreement about what to report can coexist with
17 hidden consensus about what is true. We also test whether having agents verify their
18 own analysis, via repeated cross-validation or bootstrapping, changes their reported
19 confidence. Numeric estimates tighten for genuinely robust effects, while instability
20 is exposed in fragile ones: sign agreement on a small effect fell from 90% to 70%
21 once checked. Repeating the study with a different model, Claude Haiku in place
22 of Sonnet, replicates the overall pattern. It also surfaces a new failure mode: on one
23 question, both models agreed almost completely on what to measure, yet reached
24 opposite conclusions about the value itself, a $14\times$ swing traced to an invisible
25 preprocessing choice. Finally, we independently re-executed every replicate’s
26 saved code against the original data to check our own numbers: 208/240 (87%)
27 reproduced within tolerance. Sonnet had zero contradictions between reported
28 and re-executed values. Haiku had four: cases where the agent’s self-reported
29 number did not match what its own deterministic code actually computed, a distinct
30 reliability gap we disclose in full.

31 **Keywords:** autonomous research, measure-choice diversity, nonstandard errors, LLM agents, self-
32 verification

33 1 Autonomous research result

34 This part is for the research itself: the hypothesis, method, evidence, and primary result generated or
35 developed by the agent.

1.1 Hypothesis

When independent human research teams are given the same dataset and question, they routinely reach different quantitative conclusions, not from error, but because the space of defensible analytical choices (model, controls, metric, validation scheme) is large [Steege et al., 2016, Simonsohn et al., 2020, Gelman and Loken, 2013], and different teams occupy different corners of it, rather than one team deliberately mapping the space itself. Menkveld et al. [2024] documented this directly: 164 independent teams analyzing the same financial dataset produced estimates whose disagreement (“nonstandard errors”) often rivaled or exceeded each team’s own reported statistical uncertainty. Gao and Xiao [2026] recently showed autonomous AI agents [Lu et al., 2024] reproduce this: 150 independent Claude Code agents given the same six market microstructure questions diverged sharply, driven by stable methodological forks, with AI peer review largely failing to resolve the disagreement while exemplar exposure did. Concurrently, Bertran et al. [2026] showed this AI-agent dispersion scales cheaply across observational social-science datasets, using a separate AI auditor to screen each run for validity.

All three studies are confined to observational, finance, or social-science-style empirical analysis, where the “correct” measure is inherently a matter of judgment rather than something executable and checkable, unlike machine learning research, whose central operations (train a model, evaluate it, compute a metric) are directly checkable. None manipulates question specificity or tests same-agent self-verification (Gao and Xiao rely on exemplar exposure; Bertran et al. on a separate auditor), raising two questions none address: does nonstandard-errors disagreement appear when the domain offers mechanical ground truth, and, since ML tasks are executable, does asking agents to verify their own findings via a stability check reduce disagreement the way exemplar exposure did for Gao and Xiao, given that this kind of self-check has no analogue where the underlying “truth” cannot itself be re-executed?

We hypothesize that (H-a) measure-choice diversity replicates in the ML domain, driven by whether a question’s operationalization is specified, and (H-b) self-verification changes reported numeric confidence, tightening robust effects and exposing instability in fragile ones, without reopening the measure choice, since verification runs after that choice is made. H-a predicts higher diversity for abstract than concrete questions; H-b predicts the verification arm shifts numeric dispersion/sign agreement depending on the underlying effect’s robustness.

1.2 Method

Dataset. All replicates analyze a single, static copy of the UCI/OpenML Adult (Census Income) dataset [Becker and Kohavi, 1996] (48,842 rows, mixed categorical/numeric features, binary income target, ~24% positive class), fetched once and frozen so every replicate receives byte-identical data.

Research questions. We wrote six research questions about this dataset: two abstract/concrete pairs, plus two abstract questions with an implicit method, testing whether method convergence can coexist with target divergence (Table 1; Section 1.3). Each replicate agent receives only the dataset and one research question: no documentation, no access to any other replicate’s work, and no information about which other measures might be reasonable.

Table 1: The six research questions: two abstract/concrete pairs (H1/H2, H4/H5), plus two abstract questions with an implicit method (H3, H6).

ID	Specificity	Question
H1	Abstract	Does model family meaningfully affect predictive performance?
H2	Concrete	Does default-hyperparameter random forest beat default logistic regression on stratified 5-fold CV ROC-AUC?
H3	Abstract	Which features are most important for predicting income?
H4	Abstract	Does addressing class imbalance improve model quality?
H5	Concrete	Does SMOTE oversampling change minority-class F1 by more than 0.02 versus no resampling, holding the classifier fixed?
H6	Abstract	Is the model well-calibrated?

Isolation and scale. Every replicate runs as an independent, non-interactive Claude Code CLI invocation (`claude -p ... -dangerously-skip-permissions`) in a freshly created, unpredictably-

77 named temporary directory, with an explicit instruction not to explore outside its own working
 78 directory; up to 8 replicates run concurrently, so sibling directories do coexist (Section 2.8). We
 79 collected 10 independent replicates per (hypothesis $\times$ arm) cell: 120 replicates total with Claude
 80 Sonnet, plus a matching 120-replicate cross-model check with Claude Haiku (Section 2.3).

81 **The verification arm.** Half of the replicates for each hypothesis (the “verify” arm) receive an
 82 additional instruction: before reporting, validate your finding’s stability via repeated cross-validation
 83 with different seeds, a bootstrap confidence interval, or a held-out re-test split, and report whether it
 84 held up. The other half (“no-verify”) report directly after a single analysis pass.

85 **Structured reporting.** Every replicate writes a structured result (a named metric, its value, a direction,
 86 and free-text on its methodological choices) rather than a forced common statistical form, since
 87 abstract questions admit incompatible units.

88 **Verification of our own data.** Independent of the verification *arm*, we re-executed every replicate’s
 89 saved code against the original data: 208/240 (87%) reproduced. Sonnet showed zero contradicting
 90 values; 4/120 Haiku replicates did, traced to self-reports not matching their own deterministic code
 91 (Section 2.5).

92 1.3 Evidence

93 **Measure-choice diversity is concentrated, not uniform.** Figure 1 shows the number of distinct
 94 measure framings each hypothesis produced across 20 replicates, collapsing pure wording differences
 95 (fold counts, “test set” mentions) into one canonical form (coding rule in Appendix A). The picture is
 96 uneven: H1 (abstract) produced 10 framings, entirely from disagreement over *which pair* of models
 97 to compare – the underlying metric (ROC-AUC) is uniform. H4 (abstract) produced 3, genuinely
 98 different base metrics (balanced accuracy, macro-F1, ROC-AUC). H2 (concrete) produced 1; H3
 99 and H6 – both abstract, both with an implicit method – also converge to 1: every H3 replicate uses
 100 permutation-importance ROC-AUC drop, every H6 replicate uses 10-bin ECE [Guo et al., 2017],
 101 differing only in which feature or model is named. Pooled per arm of 10, abstract questions still
 102 average more distinct measures than concrete (2.6 vs. 1.3, $2.1\times$) with a less dominant modal framing
 103 (73% vs. 98%, $1.3\times$), driven almost entirely by H1 and H4.

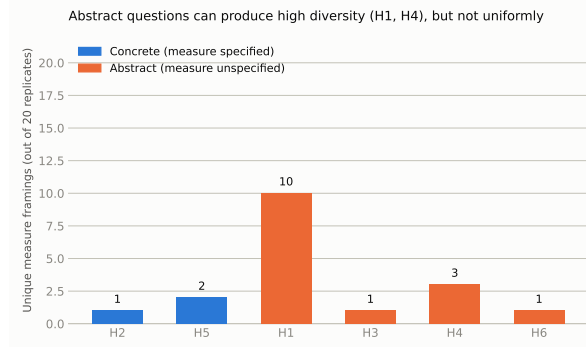


Figure 1: Unique measure framings per hypothesis (out of 20 replicates, 10 per arm). Diversity is concentrated in H1 (which model pair to compare) and H4 (which metric family); the other two abstract questions (H3, H6) converge as tightly as the concrete ones.

104 **Surface disagreement overstates substantive disagreement.** H1’s diversity count alone would
 105 suggest near-total disagreement, but that is too strong a conclusion: 20/20 replicates (100%) agreed a
 106 tree-ensemble model (Gradient Boosting [Friedman, 2001], HistGradientBoosting, or Random Forest
 107 [Breiman, 2001]) outranks linear/naive baselines, despite no majority consensus on which pair to
 108 headline (10/20 unique framings). For H4, 15/20 (75%) explicitly stated the full pattern “balanced
 109 accuracy/minority recall improves, ROC-AUC stays roughly flat”; 18/20 (90%) agreed at minimum
 110 on “improves,” despite 3 distinct base metrics.

111 **Verification changes numeric confidence, not measure choice.** Figure 2 compares the no-verify
 112 and verify arms for H2 and H5. For H2 (a robust, consistently-signed effect), sign agreement is
 113 100% in both arms, and verification tightens the spread (SD 0.0038 $\rightarrow$ 0.0022). For H5 (small and

fragile), sign agreement *falls* under verification, 90% $\rightarrow$ 70%, and the mean effect shrinks toward zero ($-0.0028 \rightarrow -0.0016$): several verify-arm replicates that saw an effect on one split found, once checked via repeated CV, that the sign was unstable across folds, and reported this explicitly. This is unlikely to be pure split-to-split randomness: all 20 replicates set `random_state=42` for their primary split, though a shared seed does not guarantee identical folds once split size, CV scheme, and preprocessing also vary; the likelier source is differing pipeline choices (imputation, encoding, SMOTE parameters), the same invisible-fork pattern as the H2 reversal below.

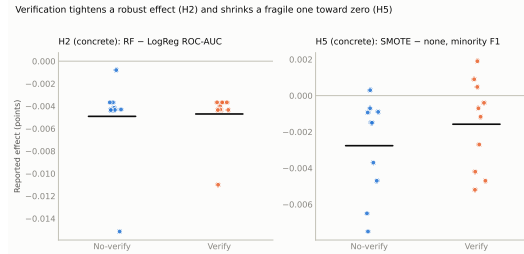


Figure 2: Replicate-level values for H2 and H5 by arm. Verification tightens a robust effect (H2) and shrinks a fragile one toward zero (H5); black bars show arm means.

A cross-model check surfaces hidden disagreement despite apparent consensus. Core patterns replicate with Haiku: framing diversity is concentrated in the same two hypotheses (Table 2), and H1’s substantive convergence holds (18/20 Haiku vs. 20/20 Sonnet agree a tree ensemble wins). H3 is the clearest case where Haiku’s extra diversity is a real methodological difference rather than wording (4 vs. 1): Sonnet always uses permutation importance, while Haiku splits across permutation, impurity-based (Gini), and multi-method consensus scores – a real difference in *method*, not just wording. More strikingly, H2 (where both models agree completely on *what* to measure, ROC-AUC difference RF–LogReg) reverses on *what value they get*: Sonnet unanimously finds logistic regression beats random forest (mean -0.0048 , both arms); 85% of Haiku replicates find the opposite, $14\times$ larger in magnitude (mean $+0.0675$), traced to an unprompted encoding difference the brief never specifies. Reading the saved code directly (self-report undercounts this): Sonnet one-hot encodes in 108/120 replicates across all hypotheses (12/120 use ordinal encoding only), Haiku uses `LabelEncoder` in 118/120, invisible in the measure name. H5, the other question where both models converge on a single measure, does *not* show this pattern: all 40 replicates across both models and arms report the SMOTE effect as below the question’s own 0.02 threshold, so this mirror-image failure mode is currently evidenced by one question, not two, and should not be over-generalized.

Table 2: Cross-model diversity: unique canonical measure framings per hypothesis (pooled, out of 20 replicates).

Hypothesis	Specificity	Sonnet	Haiku
H1	Abstract	10	9
H2	Concrete	1	1
H3	Abstract	1	4
H4	Abstract	3	3
H5	Concrete	2	2
H6	Abstract	1	2

1.4 Primary Result

Across six research questions and 240 replicates (120 Sonnet, 120 Haiku): (1) question specificity can drive large disagreement, extending nonstandard errors in AI-conducted research [Gao and Xiao, 2026, Menkveld et al., 2024] to ML, concentrated in specific questions rather than uniform across abstract wording; (2) surface disagreement overstates genuine disagreement: conclusions converge in 75–100% of cases despite near-total disagreement on what to report; (3) self-verification recalibrates confidence without touching measure choice, tightening robust effects and surfacing fragile ones; (4) core patterns replicate across a second, differently-sized Claude model, but agreement on *what* to measure does not guarantee the value: for H2, the second model reverses a unanimous finding.

2 System Design and Meta-Analysis

This is a human-author companion to the main result and should act as a meta-analysis of the research process undertaken.

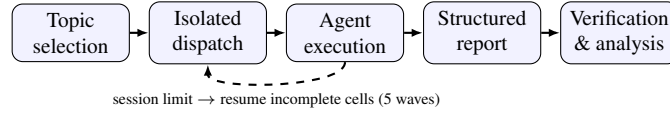


Figure 3: Research-loop pipeline (240 replicates: 120 Sonnet, 120 Haiku). The harness resumes only incomplete cells after a session-limit interruption; every replicate is re-verified before entering analysis (Section 2.5).

2.1 Agent and Harness

The primary experimental replicates were produced by Claude Sonnet [Anthropic, 2026], invoked through the standalone Claude Code CLI (@anthropic-ai/claude-code, v2.1.221) rather than through the Anthropic API directly, using the author’s existing subscription rather than separate metered billing. Each replicate is a single, non-interactive invocation: `claude -p "<brief>" -dangerously-skip-permissions -model sonnet`, run with its working directory set to a freshly created, unpredictably-named temporary directory containing only a static copy of the dataset. `-dangerously-skip-permissions` was necessary for unattended batch execution across well over a hundred replicate runs, a real reduction in interactive safety guardrails relative to normal Claude Code use. The cross-model check (Section 1.3) used an identical harness and brief template, invoked with `-model haiku` in place of `-model sonnet`.

Isolation between replicates was approximated at two levels: a fresh, non-shared filesystem location per replicate, and an explicit brief instruction not to read, list, or explore any directory other than the replicate’s own working directory. This is weaker than the container/Singularity-based isolation used in the most directly comparable prior work [Gao and Xiao, 2026], which reports having observed agents reading sibling outputs without such isolation during their own trial runs; we did not build container-level isolation for this laptop-scale study (Section 2.8).

Every replicate was asked to produce a structured result file (a named primary metric, its value, a qualitative direction, and free-text methodological-choices text) plus its analysis code as a saved script, avoiding the unit-normalization problem Gao and Xiao [2026] had to solve after data collection by asking each agent to self-report one named metric and its own reasoning rather than forcing a common statistical form up front.

2.2 Tools

The agent’s own toolset within each replicate session was the standard Claude Code toolset (file read/write, bash execution) with no additional MCP servers or external tools configured; agents wrote and executed their own Python analysis code, predominantly using scikit-learn [Pedregosa et al., 2011] (240/240), pandas [McKinney, 2010] (240/240), and numpy [Harris et al., 2020] (239/240). For the class-imbalance hypotheses, 78 of 80 H4/H5 replicates used imbalanced-learn [Lemaître et al., 2017] to apply SMOTE [Chawla et al., 2002]. Model choice shows the opposite of Section 1.3’s measure-choice pattern: Table 3 breaks down the model classes agents chose across all 240 replicates. `RandomForestClassifier` alone appears in 220/240 (92%), far ahead of `LogisticRegression` (180) and everything else. Agents converge on which tool to use even while diverging on how to report what it found.

Table 4 breaks this down per hypothesis against the measure-framing counts (Section 1.3). H2 and H5 name a model in the question, so uniformity there is not a free choice; H1, H3, and H4 leave it open and still land on `RandomForestClassifier` in all 20 replicates. H6 converges instead on `LogisticRegression` (17/20, 85%), plausibly since its outputs are already well-calibrated by construction. Model choice converges tightly even where free, just not always on the same model; measure names track a different, largely independent split (H1/H4 diverge, H3/H6 converge).

Table 3: Model classes used across all 240 replicates’ saved analysis code (not mutually exclusive; a replicate may compare several).

Model class	Replicates using it
<code>RandomForestClassifier</code>	220 (92%)
<code>LogisticRegression</code>	180 (75%)
<code>GradientBoostingClassifier</code>	45 (19%)
<code>HistGradientBoostingClassifier</code>	24 (10%)
<code>KNeighborsClassifier</code>	22 (9%)
<code>DecisionTreeClassifier</code>	21 (9%)
<code>SVC</code>	16 (7%)
<code>GaussianNB</code>	13 (5%)
<code>MLPClassifier</code>	7 (3%)
<code>AdaBoostClassifier</code>	2 (1%)

Table 4: Model-choice convergence per hypothesis (Sonnet, 20 replicates each) against unique measure framings (Figure 1). H2/H5 name a model in the question; H1/H3/H4/H6 leave it open.

Hyp.	Model named in question?	Dominant model (usage)	Unique measure framings
H1	No	<code>RandomForestClassifier</code> (20/20)	10
H2	Yes (RF vs. LR)	<code>RandomForestClassifier</code> (20/20)	1
H3	No	<code>RandomForestClassifier</code> (20/20)	1
H4	No	<code>RandomForestClassifier</code> (20/20)	3
H5	Yes (RF fixed)	<code>RandomForestClassifier</code> (20/20)	2
H6	No	<code>LogisticRegression</code> (17/20)	1

188 The orchestrating batch harness running the pipeline in Figure 3 was written in Python using
189 standard scientific-Python tooling (`concurrent.futures`, `pandas`, `scikit-learn`, `matplotlib` [Hunter,
190 2007]), run locally on a single Apple Silicon laptop with no dedicated GPU, cluster, or other cloud
191 infrastructure; timing and retry-wave details are given in Section 2.3.

192 2.3 Research Loop

193 The research loop had three phases: topic selection, data collection, and analysis.

194 **Topic selection** involved an extended, iterative novelty-checking process: roughly fifteen candidate
195 research questions across three genres (LLM agent behavior/reliability; classical ML mechanisms;
196 large-scale systematic ML audits) were proposed and checked against current literature via web
197 search, and the great majority were found to already have close prior art, in several cases published
198 within weeks of the check. The direction pursued here followed only after an earlier framing was
199 found to substantially overlap with Gao and Xiao [2026] and adjacent recent work, prompting a
200 search for a structurally distinct angle: the ML domain those studies could not reach.

201 **Data collection** proceeded in five retry waves (Figure 3) rather than a single pass, since the harness
202 resumed only cells without a completed replicate whenever the Claude Code CLI’s session usage
203 limit was hit. Collection spanned roughly 33 calendar hours, though only about 3–4 hours involved
204 any replicate actively running (the rest was idle time reflecting both the session limit and when
205 the human author was next available to relaunch collection), and the practical per-window ceiling
206 (roughly 15–56 successful heavy-agentic runs) proved far smaller and more variable than anticipated.
207 Collection also surfaced a genuine failure mode, disproportionate among compute-heavy verification-
208 arm replicates: agents would spawn their analysis as a background shell task and end their turn
209 awaiting a completion notification that could never arrive in a single-shot, non-interactive session.
210 An explicit brief instruction against this reduced but did not eliminate the pattern. A separate
211 lifecycle issue (three dispatch slots each succeeding twice under different run identifiers, traced to
212 an orchestration process halted without terminating its already-spawned subprocesses) was resolved
213 by keeping the earlier-started run of each pair, a rule fixed before any outcome was inspected. The
214 dataset was finally trimmed to a uniform $N=10$ per cell (120 total) via a fixed replicate-index range
215 set at dispatch time.

216 Table 5 extends this to Haiku: per attempt, Sonnet succeeded only 37% of the time (52% of slots
217 needed a retry) against Haiku’s 79% (8%), and ran longer on average (312s vs. 195s). Both gaps are
218 consistent with Sonnet hitting the session-limit and background-task failure modes above more often,
219 though we did not log which cause ended each failed attempt.

Table 5: Batch-dispatch attempts by model across all retry waves, from the raw dispatch log (the final dataset keeps N=10 replicates per cell, 120 per model).

Model	Attempts	Succeeded	Timed out	Replicate slots needing >1 attempt	Mean wall-clock (success)
Sonnet	363	134 (37%)	35	93/180 (52%)	312s
Haiku	152	120 (79%)	18	10/120 (8%)	195s

Analysis proceeded through iterative drafting, figure generation, and review, detailed further in Section 2.4.

2.4 Human Interventions

The human author made every scope decision throughout the loop described in Section 2.3: rejecting an initial research direction once it was found to overlap with recent prior art and directing the agent toward the ML-domain extension; choosing the execution mechanism (CLI over a raw API key) and model-family scope (single family, deferred rather than abandoned); setting and defending the replicate count (N=10 per cell) against a further reduction to N=5 the human rejected on statistical grounds, since the verification-arm comparison (this paper’s most novel manipulation) would have had too little power at that size; and reviewing drafted figures and text at each stage. During review, the human author caught a framing problem the agent had not flagged: an early draft title reused a precursor paper’s exact title as a substring, reading as a derivative reskin before a reader reached the abstract; it was changed to lead with this paper’s own contribution instead. The human author also set the boundary that all reported figures (including replicate counts and the record of process incidents in this section) must reflect what actually happened, rather than a retrospectively cleaner narrative, and that this section’s substantial drafting collaboration with the agent be disclosed rather than presented as solely human-authored, consistent with, and enforcing, this workshop’s disclosure policy.

2.5 Verification

Two distinct verification mechanisms appear in this paper and should not be conflated. The **verification arm** is an experimental manipulation under study (Section 1.3), not a claim about data quality. Separately, we independently verified the **correctness of the underlying dataset** by re-executing every replicate’s saved analysis code against the original data and comparing the reproduced primary metric to the one originally reported, a check made possible, and cheap, by the fact that ML analyses are directly executable, unlike the empirical-finance analyses in the most comparable prior work.

For the primary Sonnet dataset, 113 of 120 replicates (94%) reproduced exactly or within small numerical tolerance, and zero replicates showed a script running to completion and producing a value that contradicted its own report; every discrepancy traced to a limitation of the re-verification method itself: 6 replicates’ saved scripts were not fully self-contained (the majority cause) and 1 timed out on the heaviest verification-arm script.

Running the same check on the Haiku dataset surfaced a different and more concerning pattern: 95 of 120 replicates (79%) reproduced, and this time 4 replicates’ saved scripts ran to completion and produced a value that contradicted their own report. We manually re-ran a sample of these mismatching scripts twice each and confirmed they are fully deterministic (identical output both times), ruling out simple run-to-run non-determinism; the discrepancy is between what the saved code computes and what was self-reported, apparently produced independently rather than derived from that code’s own execution. The remaining 21 non-reproducing Haiku replicates match the same not-self-contained-script pattern seen in Sonnet, at a notably higher rate (17.5% vs. 5%). Table 6 summarizes both datasets.

We disclose all of this plainly: re-execution establishes that reported numbers were not fabricated on the runs it could fully check and confirm, but it also surfaced (rather than merely failed to rule out) a real, model-specific self-report reliability gap. This is itself consistent with this paper’s main finding: verification is not one thing, and a check aimed at one failure mode (fabrication) can, run rigorously enough, catch a different one (self-report/execution inconsistency) it was not specifically designed to find.

Table 6: Independent re-execution verification: reproduction of each replicate’s self-reported primary metric from its own saved analysis code.

Dataset	N	Reproduced	Value mismatches	Non-self-contained rate
Sonnet	120	113 (94%)	0	5%
Haiku	120	95 (79%)	4	17.5%
Combined	240	208 (87%)	4	N/A

2.6 Significance of result

The result matters for two audiences. For researchers using AI agents as empirical collaborators, it is a direct warning: an underspecified request fails quietly, not loudly, returning a confident answer to a subtly different question, and the fix (specifying the measure, not just the goal) is cheap and available today. For the autonomous-research community this workshop represents, the more important result may be that a verification step targeting numerical stability structurally cannot resolve disagreement about what to measure. As autonomous agents gain more latitude in real research pipelines, “have the agent check its own work” is a natural first line of defense; this result suggests that defense has a specific blind spot, closable only by a different intervention (in the spirit of the exemplar-exposure result of Gao and Xiao [2026]), not a more thorough version of the same one. One caveat: Section 1.1 compares self-verification to exemplar exposure, but they differ: exemplar exposure could change which measure agents pick, while our stability check runs only after that choice is made.

2.7 Interpretability

All experimental artifacts were tracked to support post-hoc audit: every replicate’s raw result, code, and metadata went to a dedicated per-run directory, and every batch dispatch and retry wave was logged, one line per attempt, so the full history is independently reconstructable rather than overwritten. Excluded and superseded runs were archived, not deleted. A separate chronological process log recording every scope decision and incident is available as supplementary material; Sections 2.3–2.4 are drawn directly from it.

Table 7 breaks the archive down by reason, from each excluded run’s saved metadata. Consistent with the background-task failure mode (Section 2.3), most excluded Sonnet attempts (194/243, 80%) never produced a result file; only 13 (5%) were extras beyond the fixed N=10 slots. The smaller Haiku archive differs: timeouts are the largest cause (16/33, 48%).

Table 7: Why archived replicate attempts were excluded, by model (from saved run metadata).

Reason	Sonnet ($N=243$)	Haiku ($N=33$)
No result file produced	194 (80%)	14 (42%)
Timed out	32 (13%)	16 (48%)
Succeeded, extra beyond N=10	13 (5%)	3 (9%)
Other error	4 (2%)	0 (0%)

2.8 Limitations

Isolation is prompt-level and filesystem-based, not OS-level sandboxing; up to 8 replicates run concurrently so siblings coexist, and we cannot rule out imperfect isolation, though found no evidence of cross-contamination. This study covers two Claude models of different capability tiers (Sonnet, Haiku), not distinct families; broader coverage (e.g. Opus, non-Anthropic) was not attempted, and Haiku’s re-verification followed initial collection rather than running alongside it. N=10 per cell, well below the 150 in the most comparable prior study, was sized for laptop/subscription-quota feasibility, not power analysis. Our re-execution tolerance ($\pm 1\%$ absolute or 5% relative) exceeds the entire effect size for H2 and H5, so a small-effect replicate can pass despite a materially different re-executed value; under a stricter same-file criterion, reproduction is 108/120 (90%) for Sonnet and 94/120 (78%) for Haiku, versus the reported 113/120 and 95/120. Measure-choice diversity and H4’s substantive-agreement figures use a documented but judgment-based coding rule (Appendix A); a different reasonable coding could shift counts by one or two.

3 Reflections and Broader Impact

This paper’s main practical implication is a warning with an inverted risk profile from the usual failure mode people worry about with autonomous agents: the danger here is not that an agent visibly fails, but that it succeeds confidently and quietly at a slightly different task than intended, in a way that is only detectable by comparing many independent attempts. Our own Sonnet-vs-Haiku encoding difference (Section 1.3) is a clean illustration: 108/120 Sonnet replicates one-hot encoded categorical features (reading the saved code, not self-report), 118/120 Haiku replicates used label encoding instead, and no replicate’s self-reported metric name hinted at the difference. As autonomous agents take on a larger share of applied ML work (model selection, feature engineering, evaluation design), this failure mode will mostly go unnoticed by a single researcher running a single analysis, since nothing about a lone run signals that a different, equally defensible choice existed. The positive implication is equally direct: this risk is cheap to mitigate where it actually bites. H1 (10 distinct framings) and its concretely-specified counterpart H2 (1 framing) are the same underlying comparison; naming the model pair, not just the metric, is what separates them, at zero additional cost. The same intervention would likely help H4 (which metric family), but does little for H3, H5, or H6, which converge on their own. We would encourage practitioners delegating open-ended ML analysis to agents to specify both the metric and the comparison target explicitly, rather than assuming the metric alone is enough.

For this workshop’s broader question of how the field should adapt, we would highlight three points beyond this paper’s specific result. First, our own novelty-checking process (roughly fifteen candidate directions checked against current literature, the great majority already covered by work published within weeks) suggests that “we searched and found nothing” is weakening as evidence of novelty specifically for autonomous research submissions, since the same tools that accelerate one group’s research loop accelerate everyone else’s. Venues may need to treat rapid, independent convergent discovery as an expected feature of the genre rather than a sign that a submission was insufficiently diligent. Second, self-verification is not one thing: this paper required two structurally different verification steps (an experimental manipulation testing whether agents checking their own numerical stability changes what they report, and an independent check that reported numbers were not fabricated), and conflating them would have been easy to do by accident. As “have the agent check its own work” becomes a standard mitigation for autonomous-agent unreliability, we would encourage reviewers and discussants to ask specifically which kind of wrongness a given verification step could and could not have caught, rather than treating any plausible-sounding check as covering all of them. Third, a laptop and a subscription are not yet a research lab, and the gap is measurable: this paper’s Sonnet collection alone needed roughly 33 calendar hours and five retry waves despite only 3–4 hours of active computation (Section 2.3), a limit imposed by an artificial usage cap, not by compute capability, and compounded by tooling reliability: Sonnet succeeded on only 37% of dispatched attempts per try, consistent with, though not conclusively caused by, the background-task completion-notification bug described above (we did not log which cause ended each failed attempt). The most comparable prior study ran 150 agents against metered billing with no such wall [Gao and Xiao, 2026]. Access to a research lab increasingly means access to a *budget* and to tooling mature enough to finish what it starts. That shapes which questions are practically askable, not just how quickly they get answered, and is worth this workshop’s attention, since it bears on who gets to participate in autonomous research at all.

This point held during our own review process, not just in the collected data: the Haiku re-verification (Section 2.5) was nearly skipped as redundant, since re-running a pipeline already applied to Sonnet seemed unlikely to surface anything new. It caught the paper’s most significant integrity finding instead, a genuine self-report/execution mismatch absent from Sonnet entirely. A check that already “passed once” feeling safe to skip on a second dataset is exactly the instinct worth resisting; that is precisely where a verification step earns its cost.

We do not see meaningful negative societal impact specific to this work: it uses a standard public benchmark dataset, introduces no new model or capability, and its practical recommendation (specify measures explicitly) reduces rather than increases risk from autonomous-agent-driven research.

A Reproducibility Statement

Three early pilot replicates (used only to validate the harness mechanics before full collection began) are excluded from every reported figure and archived separately from the 240 replicates the paper analyzes.

Measure-choice diversity coding rule. A replicate’s metric name is reduced to its base statistical measure (e.g. ROC-AUC, balanced accuracy, macro-F1, minority-class F1, Expected Calibration Error, or a feature-importance sub-type: permutation, impurity/Gini, or multi-method consensus), stripping wording that does not change what is computed (fold/CV/test-set phrasing, which specific feature or model is named). For H1 only, which pair of model families is compared is kept as part of the code, since the question “does model family affect performance” is not well-defined without choosing a pair; for H3’s “top feature” and H6’s “which model to calibrate,” the named entity is an output of a fixed method rather than a chosen operationalization, so it is not counted (model choice is reported separately, Table 3/4). This is a documented, but ultimately judgment-based, coding decision; the exact rule is implemented in `harness/analyze.py` (`base_metric` and `h1_comparison_target`) and is re-run, not hand-edited, to produce every diversity number in this paper. The H4 substantive-agreement figures (Section 1.3) use a similarly documented rule: a replicate counts toward the full “improves, ROC-AUC flat” pattern only if it explicitly states both clauses, and toward “improves” alone if it states any accuracy/recall-family improvement, even without an explicit ROC-AUC clause.

The dataset (UCI/OpenML Adult/Census Income [Becker and Kohavi, 1996]) is public and freely available; no dataset-level documentation in the style of Geburu et al. [2021] accompanies it beyond the UCI repository page, which is itself disclosed to replicate agents as the only provenance information they receive. The six research questions are given in full in Table 1; the exact brief text given to each replicate agent is available as supplementary material (`harness/hypotheses.py`) and has five parts: (1) role framing as an independent, single-shot researcher with no access to other replicates’ work; (2) the dataset and its row count, with no further documentation; (3) the one research question this replicate receives, plus an explicit instruction to make its own methodological choices (named examples: preprocessing, *encoding*, split, hyperparameters, metric) for anything the question leaves unspecified; (4) an explicit instruction to run everything synchronously in the foreground rather than backgrounding long-running commands, since no follow-up turn will arrive (Section 2.3); and (5) a required `result.json` schema (summary, named metric, value, direction, methodological choices), extended with two verification fields for the verify arm. Reproducing the replicate-collection process requires the Claude Code CLI and access to Claude Sonnet and Claude Haiku under the same or comparable versions; because model behavior can drift across versions, we do not expect bit-exact reproduction of individual replicate outputs, but do expect the qualitative pattern (concentrated, question-specific measure-choice diversity rather than a uniform abstract-vs-concrete split; verification tightening robust effects and surfacing fragile ones) to be checkable by any lab with CLI access to a comparable agent. All analysis code (`harness`, verification, figure generation) is standard Python/scikit-learn and is included in supplementary material along with the full experimental log described in Section 2.3 and a fuller set of representative replicate examples than Appendix C shows. *[Author note: confirm final code/data hosting location (e.g., an anonymized repository) before submission, and update this statement with the access details.]*

B Disclosure Statement

Agent(s) / model(s) used Claude Sonnet and Claude Haiku [Anthropic, 2026], via the Claude Code CLI (@anthropic-ai/claude-code v2.1.221); Anthropic’s public model documentation serves the disclosure role model cards [Mitchell et al., 2019] are designed for, though it is not published in that specific format. The agent conducted all 240 experimental replicates (120 Sonnet, 120 Haiku: data analysis, code writing and execution, result reporting) as described in Section 1–2, and substantially assisted in drafting this paper, including Section 1 in full and a first draft of Section 2, subsequently reviewed, corrected, and revised by the human author as described in Section 2.4.

☐ A human author has curated this work and verified every claim, figure, methodology detail, and reference.

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406 verified artifact.

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408 reference below; do not check these before that review is actually done.]

409 C Representative Examples

410 Table 8 shows the raw (uncanonicalized) reported metric from five no-verify replicates of H1 (abstract)
411 and H2 (concrete): measure-choice diversity at the replicate level, not just aggregate counts. H2’s
412 five rows differ only in CV-related wording, not in substance, illustrating why the canonicalization in
413 Section 1.3 is necessary before counting. Full reasoning and saved code are in the supplementary
414 material.

Table 8: Representative examples: reported metric per replicate. H1 is abstract, H2 is concrete.

Hyp.	Replicate	Reported metric
H1	000	ROC-AUC difference (GradientBoosting—LogisticRegression), 5-fold CV
H1	001	ROC-AUC gap (HistGradientBoosting—KNN), 5-fold CV mean
H1	002	Test ROC-AUC range (HistGradientBoosting—GaussianNB)
H1	003	CV ROC-AUC spread (HistGradientBoosting—KNN)
H1	004	Test ROC-AUC spread (best—worst model family)
H2	000	ROC-AUC difference (RF—LogReg), mean over 5 folds
H2	001	ROC-AUC difference (RF—LogReg)
H2	002	ROC-AUC difference (RF—LogReg), mean over 5 stratified CV folds
H2	003	ROC-AUC difference (RF—LogReg)
H2	004	ROC-AUC difference (RF—LogReg)

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Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: This research involves no crowdsourcing and no human subjects; all experimental replicates are AI agents analyzing a pre-existing public dataset.

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- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
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